

1. Introduction to Plant Physiology in Propagation

To master air layering, you need to understand what's happening inside the plant. Marcotting works not just because of technique—but because of how plants naturally transport nutrients and respond to injury.

In this module, we explore:

How plants move water and sugars

Why roots form at the girdled site

The biological triggers behind successful propagation

2. The Vascular System: Xylem and Phloem

Plants rely on two critical transport systems:

Xylem (Upward Transport)

Moves water and minerals from roots to leaves

Located deeper in the stem

Remains intact during air layering

Phloem (Downward Transport)

Moves sugars and nutrients from leaves to roots

Located just beneath the bark

This is what you remove during girdling

Key Concept

When you remove the bark (phloem), sugars accumulate above the cut, creating the perfect environment for root development.

3. Why Girdling Works

Girdling interrupts the plant's normal nutrient flow.

What Happens After Girdling?

Sugars produced by photosynthesis cannot move downward

These sugars accumulate at the wound site

Hormones concentrate in that area

The plant responds by producing adventitious roots

Important Insight

Air layering does not create roots randomly—it redirects the plant's natural healing response into root production.

4. Role of Plant Hormones (Auxins)

What Are Auxins?

Auxins are plant hormones responsible for:

Cell elongation

Root initiation

Growth direction

In Marcotting

Auxins naturally accumulate above the girdle

Applying rooting hormone boosts this effect

Common Rooting Hormones

IBA (Indole-3-butyric acid)

NAA (Naphthaleneacetic acid)

Pro Insight

Even without synthetic hormones, plants can root—but adding them:

Speeds up rooting

Produces stronger root systems

Increases success rate

5. Adventitious Root Formation

Definition

Adventitious roots are roots that form from non-root tissues (like stems or branches).

Stages of Root Development

Wound Response

Cells at the cut site begin dividing

Callus Formation

A mass of undifferentiated cells forms

Root Initiation

Root primordia (early root structures) develop

Root Emergence

Roots grow outward into the medium

6. Environmental Factors Affecting Rooting

Successful marcotting depends heavily on the environment.

Moisture

Must remain consistently moist

Too dry = roots fail

Too wet = rot and fungal growth

Temperature

Ideal range: 20°C – 30°C

Warmer temperatures speed up root formation

Light

Indirect light is best

Too much sunlight can dry out the medium

Oxygen

Roots need air

This is why compact soil is avoided

7. Choosing the Right Timing

Timing significantly impacts success.

Best Time to Marcot

Active growing season

Typically spring to early rainy season

Avoid

Dormant periods

Extreme heat or drought

Heavy storms (can damage wrapped layers)

8. Common Mistakes in Root Formation

Mistake 1: Incomplete Cambium Removal

Result: Bark heals instead of rooting

Mistake 2: Dry Medium

Result: No root development

Mistake 3: Overwatering

Result: Rot and fungal infection

Mistake 4: Weak Branch Selection

Result: Poor or slow rooting

9. Demonstration Scenario

Example: Marcotting a Mango Branch

After girdling, sugars accumulate above the cut

Within 2–3 weeks, callus tissue forms

By week 4–6, visible white roots appear

By week 6–8, roots mature and strengthen

This demonstrates how physiology directly translates into practical results.

10. Module Summary

By the end of this module, you should understand:

How the plant's vascular system enables air layering

Why girdling triggers root formation

The role of hormones in rooting

The stages of adventitious root development

Environmental conditions required for success